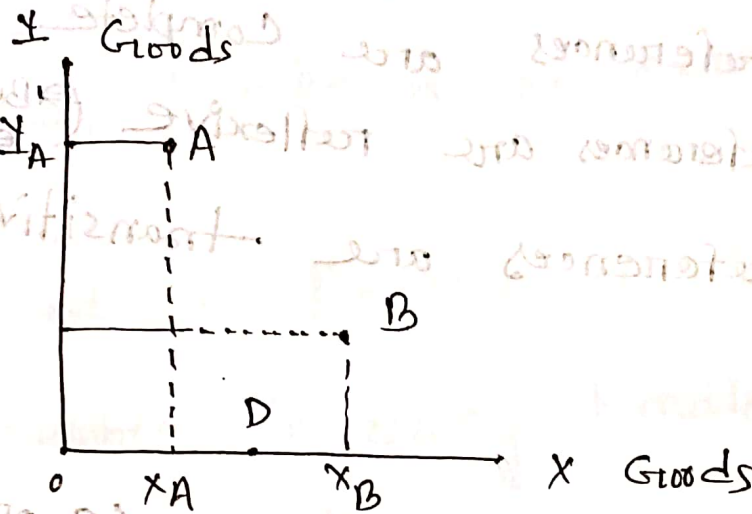


Preferences and Utility :-

X and Y are goods (things you want)

$D \rightarrow$ a bundle with only one good

A & B have same of both



Axioms on Consumer Preferences :-

Preference ordering $A \succ B \rightarrow A$ is strictly preferred to B

$A \geq B \rightarrow$ This person likes A at least as much as B .

If $A \geq B$ & $B \geq A \rightarrow$ You are indifferent

Notation:-

$A^P B \rightarrow$ A preferred to B

$A^I B \rightarrow$ A indifferent to B

Axioms:-

A1 $\rightarrow$ Preferences are complete

A2 $\rightarrow$ Preferences are reflexive (Bundle should be not empty)

A3 $\rightarrow$ Preferences are transitive

A1:

All bundles are in either Φ or Φ_j

$\downarrow$
 Φ

$\downarrow$
 Φ_j

A2: Φ is non-empty

A3: If $A > B$ and $B > C$ then $A > C$

Φ and Φ_j are disjoint (Bundle cannot be in both Φ & Φ_j)

{disjoint $\Rightarrow$ Either bundle Φ or Φ_j } $\{ \Phi \cup \Phi_j = S \}$

Let assume, A3 holds,

Given A1 — A3, preferences can be represented in ~~an~~² ways.

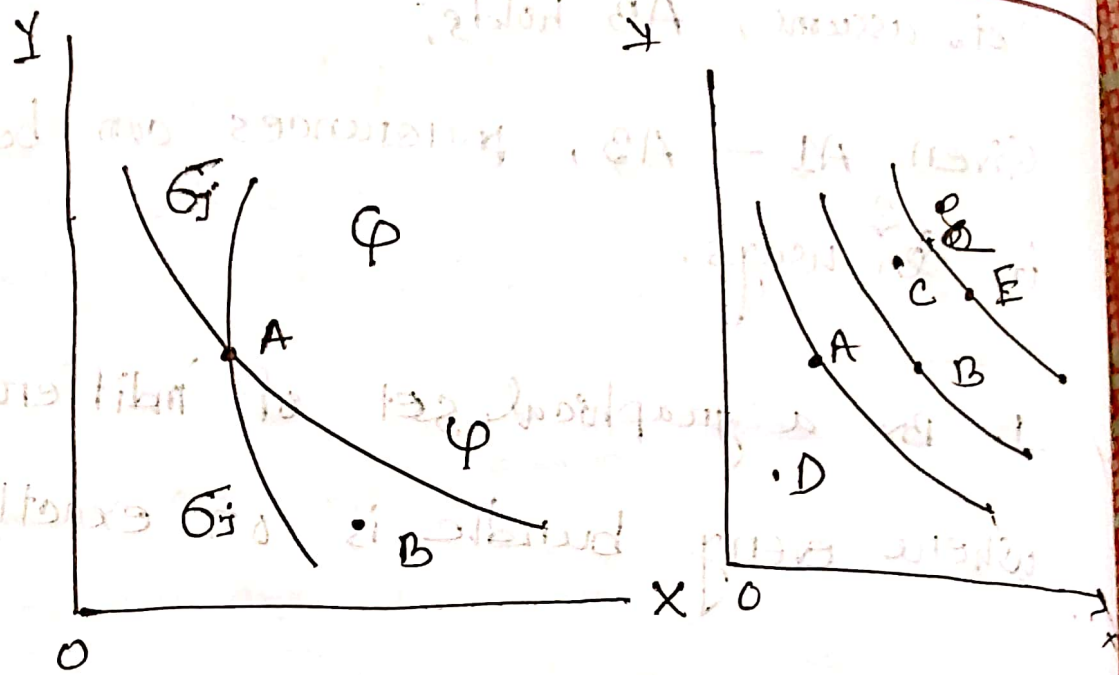
1. By a graphical set of indifference curves where every bundle is on exactly one curve.

2. By a utility function $U(\cdot)$ that assigns only a number to every bundle. The number assigned to bundle A is greater than the number assigned to bundle B if and only if $A \succ B$.

$$\varphi = \{x \in X : x \succeq A\} \Rightarrow \text{Bundle-1}$$

Note: by A2, φ is not empty

$$\sigma = \{x \in X : A \succ x\} \Rightarrow \text{Bundle-2}$$



Key:- Every bundle is on exactly one curve
(i.e., it, can't cross)

φ1: Preferences are continuous

if $A \succ B$ and $C \rightarrow B$ then $A \succeq C$

"Smooth I_c "

φ2: Preferences are monotonic (exhibit local nonsatiation)
 "more is better"

if A is B plus something
 $A \succ B$

$\exists$ — there exist only one

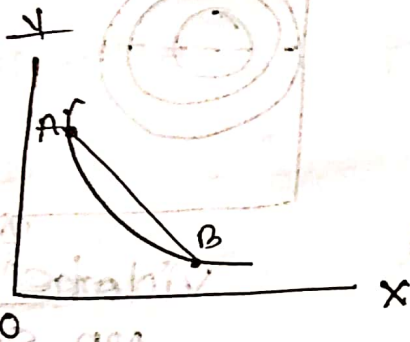
If $x_A^i \geq x_B^i \quad \forall i \rightarrow$ for all or for any

and $x_A^j \geq x_B^j$ for at least one j .

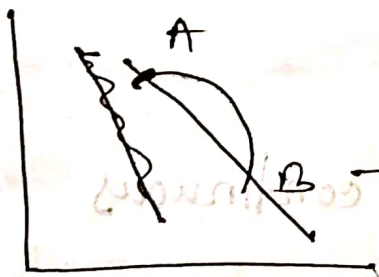
Q3: Preferences are convex

If $A \succ C$ and $B \succ C$ then

$$\lambda \in [0, 1] \quad \lambda A + (1 - \lambda) B \succ C$$

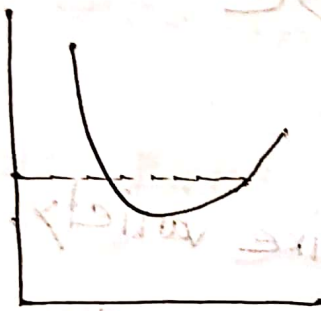
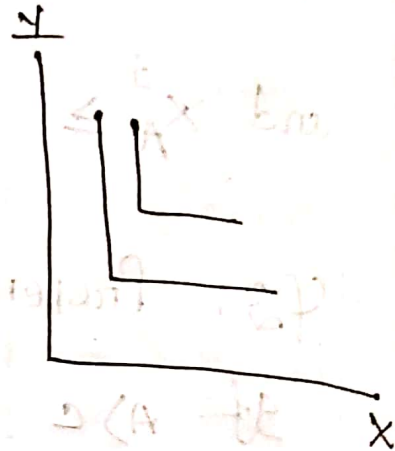
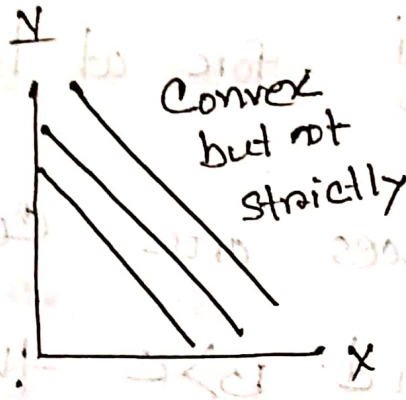
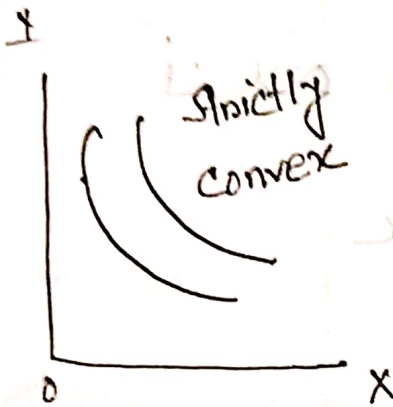


$\rightarrow$ If you like variety,
Your preferences are convex

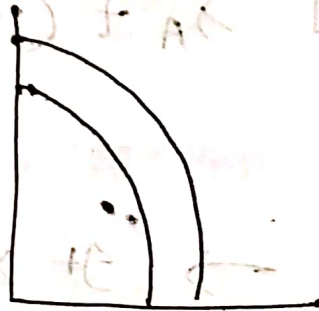


$\rightarrow$ If not like variety

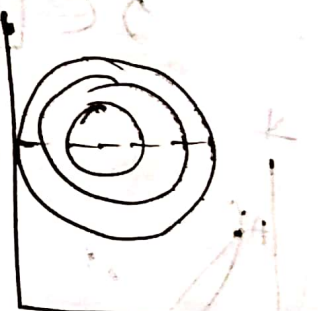
Given $A_1 - A_3$ and $\phi_1 - \phi_3$



Violates ϕ_1 : Local non satiation



Violates ϕ_3



Violates ϕ_2 local non satiation



Violates $\phi_2 \rightarrow$ Preferences are continuous

Preferences: Convex Preferences / IC

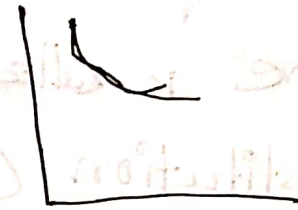
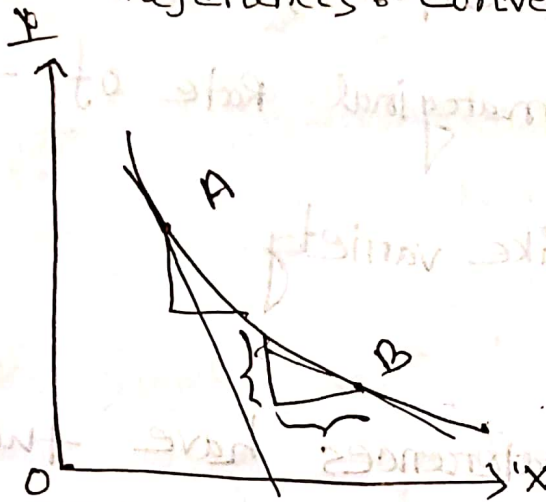


fig. Not strictly convex

fig. smooth

Slope of IC: $\frac{\Delta Y}{\Delta X}$ while staying as happy as before.
|slope| = Smallest amount of good Y you are willing to accept in a trade for one unit of good X.

OR, The person's relative valuation of X in terms of Y with strictly convex preferences, how you value X in terms of Y will depend on how much X & Y you have.

⇒ Initially we willing to sacrifice, but $\frac{\Delta Y}{\Delta X}$ (slope of IC)
Next $\frac{\Delta Y}{\Delta X}$ represent Neg. Sloped = MRS (slope of IC)

The negative of the slope of the indifference curve is called the marginal rate of Substitution (MRS) $\rightarrow$ Like variety

MRS (x, y): Convex preferences have the property of diminishing MRS $\frac{dMRS}{dx} < 0$

A1 - A3:

Allow us to represent preferences by a utility

function $U(A)$

Write the property

$U(A) > U(B)$ iff $A \succ B$

$U(A) = U(B)$ iff $A \sim B$

Typically have 2 goods (x, y)

$U(A) = U(x_A, y_A)$

$U(B) = U(x_B, y_B)$

How to get MRS from $U: U(x, y)$

Along an indifference curve, $U(x, y) = \text{constant}$
we need $-\frac{dy}{dx} \big|_{u=\bar{u}}$

Take total derivatives

$$dU = \frac{\partial U}{\partial x} \cdot dx + \frac{\partial U}{\partial y} \cdot dy$$

but on indifference curve $dU = 0$

$$\frac{\partial U}{\partial x} \cdot dx + \frac{\partial U}{\partial y} \cdot dy = 0$$

$$\Rightarrow \frac{\partial U}{\partial x} \cdot dx = - \frac{\partial U}{\partial y} \cdot dy$$

$$\Rightarrow - \frac{dy}{dx} \big|_{dU=0} = \frac{\partial U / \partial x}{\partial U / \partial y}$$

$$= MRS_{x/y}$$

$\Rightarrow$ Proof MRS is nothing but slope

$\Rightarrow$ which is negatively slope

$\Rightarrow$ it negative at every point

$\Rightarrow$ also negatively diminishing (How strong MRS)

why diminishing?
Ans $\Rightarrow$

$\frac{\partial U}{\partial x}$ = Marginal utility of x, MU_x

$\frac{\partial U}{\partial y}$ = " " " " MU_y

Q2 get $\frac{\partial U}{\partial x} > 0$

Preferences are monotonic:-

More is better

Diminishing MRS = $\frac{dMRS}{dx} \Big|_{u=\bar{u}} < 0$

$\frac{dMRS}{dx} \neq \frac{\partial MRS}{\partial x}$ along on indifference curve

$$dMRS = \frac{\partial MRS}{\partial x} dx + \frac{\partial MRS}{\partial y} dy$$

$$MRS = \frac{U_x}{U_y} \quad \frac{\partial MRS}{\partial x} = \frac{U_{xx} U_y - U_{xy} U_x}{U_y^2}$$

$$dMRS = \frac{\partial MRS}{\partial x} dx + \frac{\partial MRS}{\partial y} dy$$

want, $\frac{dMRS}{dx} \Rightarrow \frac{dMRS}{dx} = \frac{\partial MRS}{\partial x} + \frac{\partial MRS}{\partial y} \frac{dy}{dx}$

[divided by dx]

$$\frac{dMRS}{dx} = \frac{\partial MRS}{\partial x} - MRS \left[\frac{\partial MRS}{\partial y} \right]$$

$$\frac{dMRS}{dx} = -MRS \left[\frac{\partial MRS}{\partial y} \right] < 0$$

$$\Rightarrow \frac{\partial MRS}{\partial x} = 0$$

a f.o.c

$$U = x^a y^b$$

□ Cobb-Douglas:

$$U = x^a y^b \quad a, b > 0$$

$$MRS = \frac{U_x}{U_y} = \frac{a x^{a-1} y^b}{x^a b y^{b-1}} = \frac{a}{b} \cdot \frac{y}{x}$$

$$MRS = \frac{a}{b} \cdot \frac{y}{x}$$

$$\frac{\partial MRS}{\partial x} = -\frac{a}{b} \cdot \frac{y}{x^2}$$

$$\frac{\partial MRS}{\partial y} = \frac{a}{bx}$$

Now,

$$\frac{dMRS}{dx} = \frac{\partial MRS}{\partial x} + \frac{\partial MRS}{\partial y} \cdot \frac{dy}{dx}$$

$$\frac{dMRS}{dx} = \frac{\partial MRS}{\partial x} + \frac{\partial MRS}{\partial y} \cdot \frac{dy}{dx}$$

$$= -\frac{a}{b} \cdot \frac{y}{x^2} - \left[\frac{a}{b} \cdot \frac{y}{x} \right] \left(\frac{a}{bx} \right)$$

$$= -\frac{a}{b} \cdot \frac{y}{x^2} - \frac{a^2}{b^2} \cdot \frac{y}{x^2}$$

$$> -\frac{ay}{bx^2} \left(1 - \frac{a}{b} \right) < 0$$

Quasi-Linear Preferences:-

$$V = x + 2\sqrt{y}$$

$V = x + f(y)$ where $f(y)$ is concave

$$MRS_{xy} = \frac{V_x}{V_y} = \frac{1}{y^{-1/2}} = \sqrt{y}$$

$$\frac{dMRS}{dx} = \frac{\delta MRS}{\delta x} - MRS \frac{\delta MRS}{\delta y}$$

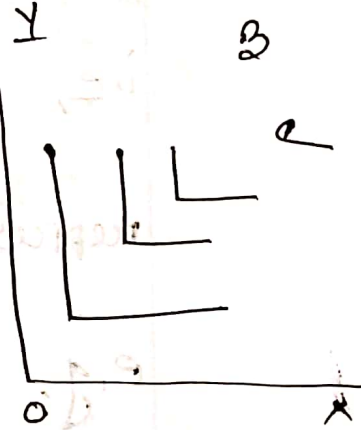
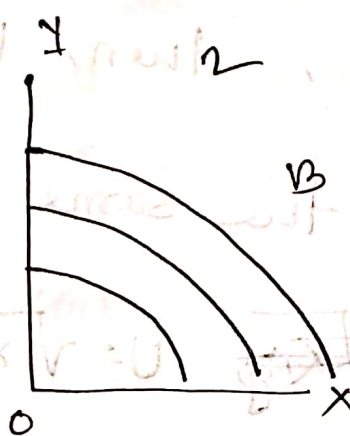
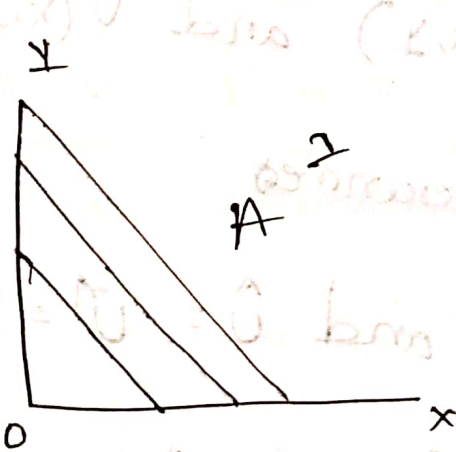
$$= 0 - \sqrt{y} \left[\frac{1}{2\sqrt{y}} \right]$$

$$= -\frac{1}{2} < 0$$

1. $U = ax + by$

2. $U = \min(ax, by)$

3. $U = x^2 + y^2$



$$\frac{\frac{\partial U}{\partial y}}{\frac{\partial U}{\partial x}} = \frac{\partial U}{\partial x} \quad \text{MRS}$$

$$\frac{bx/ax}{bx/by} = \frac{by}{bx}$$

$$\text{MRS} = \frac{bx/ax}{bx/by} = \frac{by}{bx}$$

What happens if we transform the utility function

$$\hat{U}(x, y) = f(U(x, y))$$

if f is a positive monotonic ~~function~~ transformation

i.e., $f' > 0$, then, $\hat{U}(x, y)$ and $U(x, y)$

represent the same preferences.

e.g., ~~$U = \sqrt{x, y}$~~ $U = \sqrt{xy}$ and $\hat{U} = U^4 = x^2 y^2$

same

$$\text{Let's find } \hat{MRS} = \frac{\partial \hat{U}}{\partial x} / \frac{\partial \hat{U}}{\partial y}$$

$$= \frac{\partial f / \partial U}{\partial f / \partial U} \cdot \frac{\partial U / \partial x}{\partial U / \partial y}$$

$$= \frac{\partial U / \partial x}{\partial U / \partial y} = MRS$$

Suppose $U(A) = 100$ $U(B) = 1$ and $U(C) = 0.99$

All you can say is $A > B$; $B > C$

Cannot correctly say, "This person likes A a lot more than B, but if $U(A+B) = U(A) + U(B)$ is close to indifferent between B and C

$$U(A+B) \neq U(A) + U(B)$$

Cannot compare one person's utility to other.

$U_A(x, y) > U_B(x, y)$ cannot say A likes then bundle more than person B. Can compare MRS's across people.

if $U(A) = 5$

$U(B) = 1$

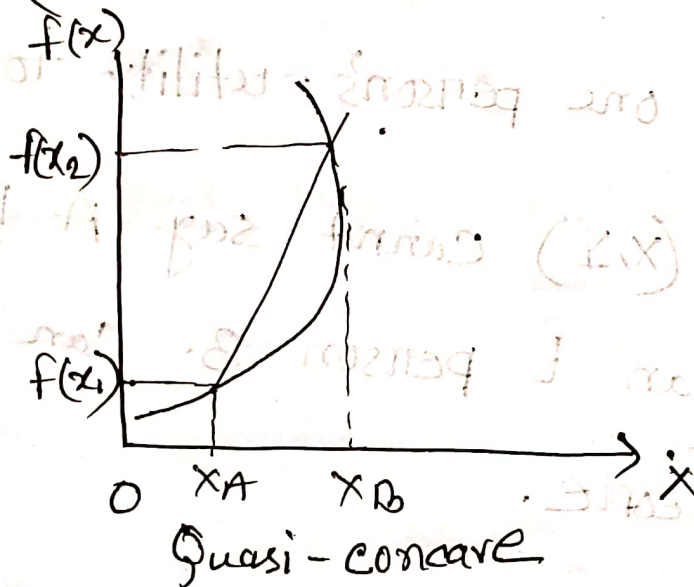
Cannot say A times more than person B

Strictly convex: Two ways to show an indifference curve is strictly convex

$\frac{dMRS}{dx} < 0$ or show $U(\cdot)$ is strictly quasi-

All quasi-concave functions have convex level

sets (i.e., indifference curve).



Concavity $\rightarrow$ look at Hessian Matrix H . If H is negative definite then the function is strictly concave.

For quasi-concave we use bordered Hessian

$$U = U(x, y)$$

$$H = \begin{bmatrix} U_{xx} & U_{xy} \\ U_{yx} & U_{yy} \end{bmatrix}$$

$$BH = \begin{bmatrix} 0 & U_x & U_y \\ U_x & U_{xx} & U_{xy} \\ U_y & U_{yx} & U_{yy} \end{bmatrix}$$

Preferences are not the only thing that ^{determines} ~~differs~~ what we buy, what we can actually buy makes up our choice set,

What limits our choices

* Income: How much we can spend, budget

* Legal

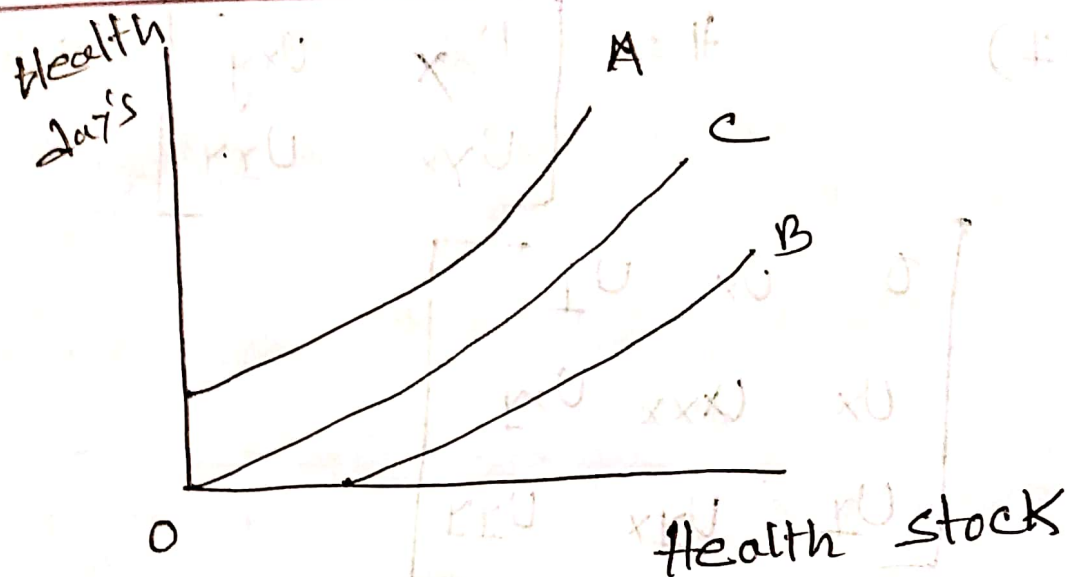
* Moral/Ethical/religious

* Health/dietary

* Time

Subsistence constraints

→ Minimum amount of calories to survive
→ Mealtimes



* Constrained optimization Problem

We will work primarily with the budget constraint,

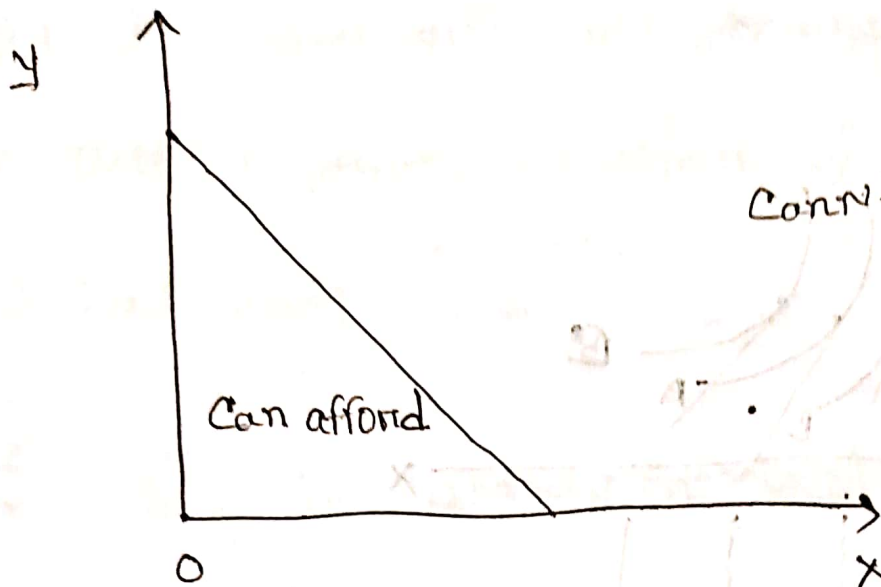
2 Goods x = How much we buy of good x
 y = " " " " " " y

P_x = price of unit of good x

P_y = " " " " " " y

I = budget (income)

$$P_x \cdot x + P_y \cdot y \leq I$$



Can't afford,

Multiple Goods

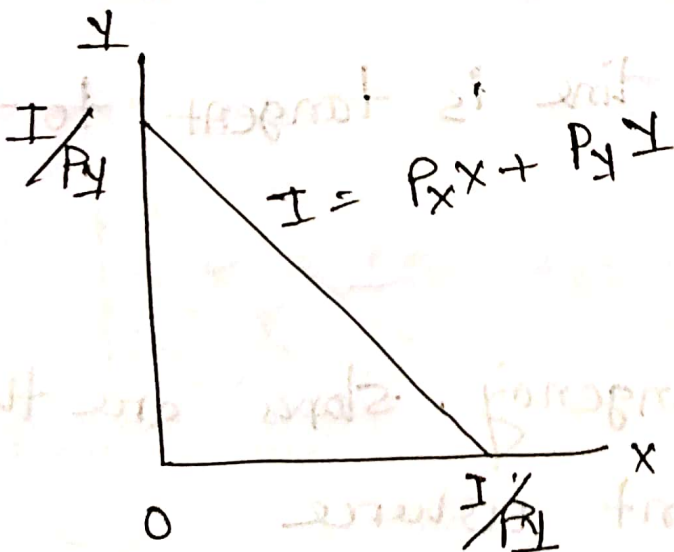
$$\sum_{i=1}^n P_i X_i \leq I$$

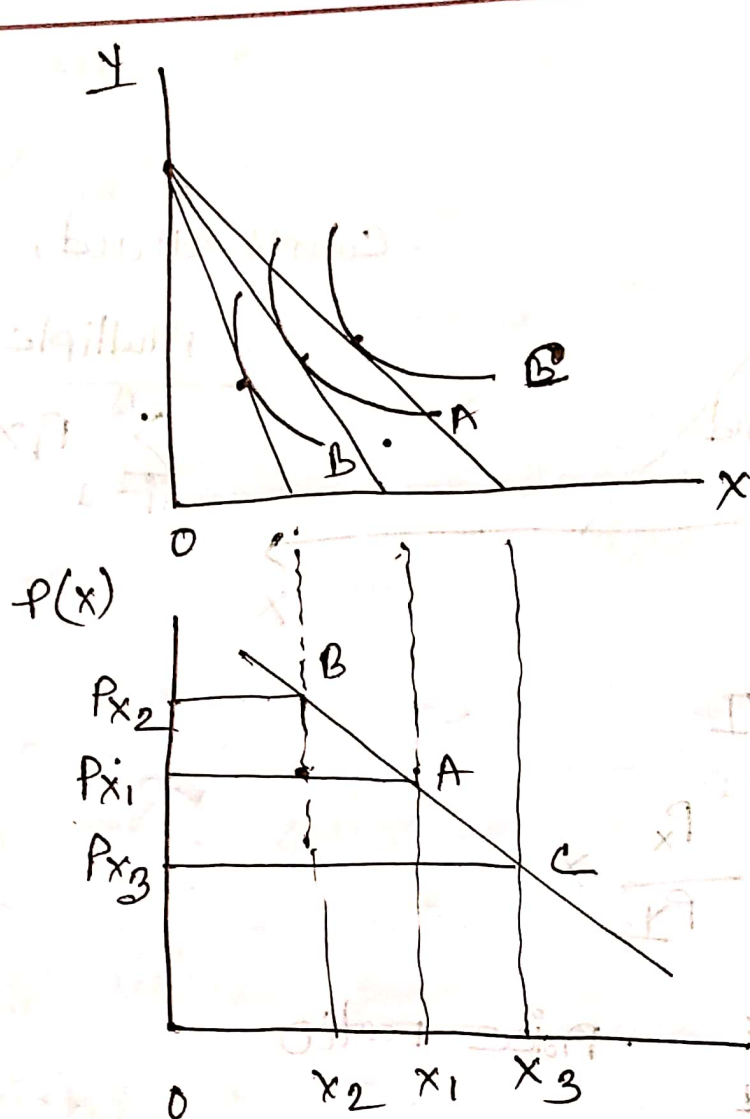
$$P_x X + P_y Y = I$$

$$Y = \frac{I}{P_y} - \frac{P_x}{P_y} \cdot X$$

$$\text{slope} = - \frac{P_x}{P_y} \cdot \text{Price ratio}$$

How the market values x in terms of y





Note: The optimal bundle is at a place where the budget line is tangent to the I.

Tangency Condition:-

At the point of tangency, slopes are the same.
Budget line Indifferent where

$$-\frac{P_x}{P_y} = -MRS$$

We get the most out of trading with the market when our relative valuation of x in terms of y equals the market

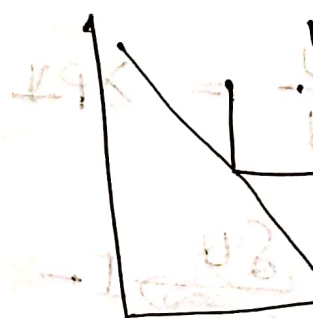
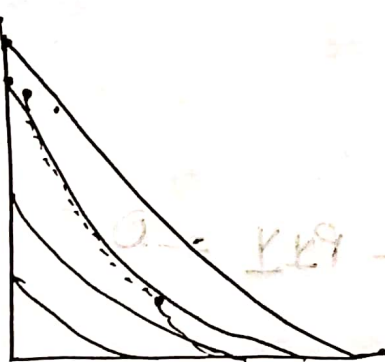
Catch: Tangency condition will only work if we have diminishing MRS

$$U = x^2 + y^2 \rightarrow \text{MRS} \uparrow$$

$$U = ax + by \rightarrow \text{MRS (NO)} - \text{can't get tangency}$$

$$U = x + f(y) \rightarrow f'' < 0$$

$$U = \min(ax, by) \rightarrow$$



$U = \min(ax, by)$
Can't get ~~slopes equal~~
~~no~~

slope must be
at the "point"

regardless of prices,
Point in where
 $ax = by$

Let's use, math econ tools,

$$\text{Max}_{x,y} U = U(x,y)$$

$$\text{s.t.} = P_x x + P_y y = I$$

$$\text{Max}_{x,y,\lambda} \mathcal{L} = U(x,y) + \lambda [I - P_x x - P_y y]$$

$$\text{3 FOC } \frac{\partial \mathcal{L}}{\partial x} = \frac{\partial U}{\partial x} - \lambda P_x = 0$$

$$\Rightarrow \frac{\partial U}{\partial x} = \lambda P_x$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{\partial U}{\partial y} - \lambda P_y = 0$$

$$\Rightarrow \frac{\partial U}{\partial y} = \lambda P_y$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - P_x x - P_y y = 0$$

$$\frac{\partial U / \partial x}{\partial U / \partial y} = \frac{P_x}{P_y} = \text{MRS}$$

$$U = XY$$

$$\text{Max } \mathcal{L} = XY + \lambda [I - P_X X - P_Y Y]$$

$$X, Y, \lambda$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial X} &= Y - \lambda P_X = 0 \Rightarrow Y = \lambda P_X \\ \frac{\partial \mathcal{L}}{\partial Y} &= X - \lambda P_Y = 0 \Rightarrow X = \lambda P_Y \end{aligned} \quad \left\{ \begin{aligned} \frac{X}{Y} &= \frac{P_X}{P_Y} \\ \Rightarrow Y &= \frac{P_X X}{P_Y} \\ \Rightarrow Y &= \frac{P_X \cdot \frac{I}{2P_X}}{P_Y} \end{aligned} \right.$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - P_X X - P_Y Y = 0$$

$$\Rightarrow I = P_X X + P_Y Y$$

$$\Rightarrow I = P_X X + \cancel{P_Y} \frac{P_X X}{\cancel{P_Y}}$$

$$= 2 P_X X$$

$$\Rightarrow X^* = \frac{I}{2 P_X}$$

Marshallian demand function

$$\text{for } X^* = \frac{I}{2 P_X}$$

$$Y^* = \frac{I}{2 P_Y}$$

Quasi-linear :-

$$U = x + 2\sqrt{Y}$$

Max x, Y, λ , $\mathcal{L} = x + 2\sqrt{Y} + \lambda [1 - P_x x - P_y Y]$

$$\frac{\partial \mathcal{L}}{\partial x} = 1 - \lambda P_x = 0 \Rightarrow 1 = \lambda P_x$$

$$\frac{\partial \mathcal{L}}{\partial Y} = \frac{1}{\sqrt{Y}} - \lambda P_y = 0 \Rightarrow \frac{1}{\sqrt{Y}} = \lambda P_y$$

$$\left(\begin{array}{l} \frac{1}{\sqrt{Y}} = \lambda P_y \\ \frac{1}{\sqrt{Y}} = \lambda P_y \end{array} \right) \Rightarrow \sqrt{Y} = \frac{P_x}{P_y} \Rightarrow Y^* = \left(\frac{P_x}{P_y} \right)^2$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 1 - P_x x - P_y Y = 0$$

$$\Rightarrow P_x x = 1 - P_y Y$$

$$\Rightarrow P_x x = 1 - P_y \cdot \frac{P_x^2}{P_y \cdot P_y}$$

$$\Rightarrow P_x x = 1 - \frac{P_x^2}{P_y}$$

$$\Rightarrow x = \frac{1}{P_x} - \frac{P_x \cdot P_x}{P_y \cdot P_x}$$

$$\therefore x^* = \frac{1}{P_y} - \frac{P_x}{P_y}$$

These are ~~correct~~ ^{correct} for $I \geq \frac{P_x^2}{P_y}$

sf $I < \frac{P_x^2}{P_y}$, $x^* = 0$ $y^* = \frac{I}{P_y}$

Ex $U = x + y$, $P_x = 2$ $P_y = 3$ $I = 300$

Demands:

$x^* = \frac{I}{P_x}$ for $P_x < P_y$

$x^* = 0$ for $P_x > P_y$

$y^* = \frac{I}{P_y}$ for $P_x > P_y$

$y = 0$ for $P_x < P_y$

Max

$U(x, y, \lambda) = x + y + \lambda [300 - 2x - 3y]$

$\frac{\partial L}{\partial x} = 1 - 2\lambda = 0 \Rightarrow 1 = 2\lambda$

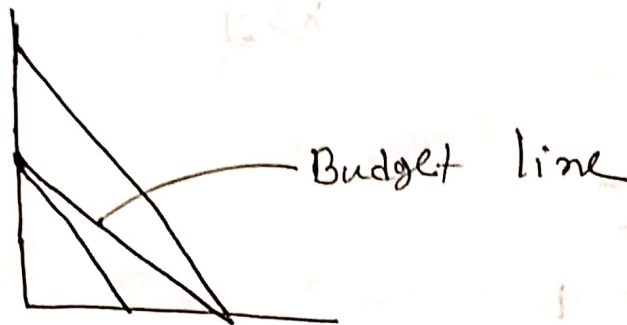
$\frac{\partial L}{\partial y} = 1 - 3\lambda = 0 \Rightarrow 1 = 3\lambda$

$\lambda = \frac{2}{3}$

$\frac{\partial L}{\partial \lambda} = 300 - 2x - 3y = 0$

Corner solution

Budget line flatter than indifference curve



$$MRS = I$$

$$P_x / P_y = \frac{2}{3}$$

Since $P_y = 2$ $P_x = 3$

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} \cdot \frac{MU_x}{P_x} = \frac{1}{2} < \frac{1}{3} = \frac{MU_y}{P_y}$$

Only buy x

Qs:- एकल commodity की मांग Purchase करवा
note आसानी से लिख दिते हैं।

Qs:- Cob-Douglas: $U = XY$

$$MU_x = Y$$

$$MU_y = X$$

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

$$\frac{y}{p_x} = \frac{x}{p_y}$$

$$y > x$$

$$x > y$$

$$U = x + 2\sqrt{y}$$

$$\frac{MU_x}{p_x} = \frac{1}{p_x}$$

$$\therefore \frac{MU_y}{p_y} = \frac{2\sqrt{y}}{p_y}$$

Buy ~~whichever~~ whichever

$$U_{\min}(a_x, b_y)$$

$$MU_x = a \quad \text{if } a_x < b_y$$

$$= 0 \quad \text{if } a_x \geq b_y$$

There is no MRS

Tangency always at the corner i.e. where

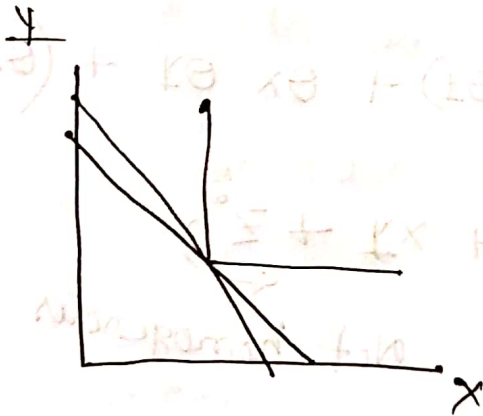
$$a_x = b_y \quad a_x = b_y$$

$$y = \frac{a_x}{b}$$

$$p_x x + p_y y = I$$

$$p_x x + p_y \cdot \frac{a_x}{b} = I$$

$$x \left[P_x + P_y \cdot \frac{a}{b} \right] = I \quad \left| \quad x = \frac{I}{P_x + P_y \cdot \frac{a}{b}} \right.$$



Homogeneity:-

Multiply all price and income by same 
Constant

$$\frac{\partial I}{\partial P_y} = \frac{I}{P_y} \cdot \frac{P_y}{P_x + P_y \cdot \frac{a}{b}}$$

$$\Rightarrow \frac{\partial I}{\partial P_x} = \frac{I}{P_x}$$

if a function consists of a bunch of variables
variables multiplies together
it is a homogenous of the degree that the
exponents add to.

$$A. \quad U = x^2 y^2 = \theta^3 x y + \theta^2 x y + \theta^3 z$$

$$B. \quad x = \frac{1}{p_x} = (\theta x)^2 (\theta y) + \theta x \theta y + (\theta z)^3$$

$$C. \quad U(x, y, z) = x^2 y + x y + z^3$$

Not homogenous

$$D. \quad U = xy + z \rightarrow \text{Not homogenous}$$

$$U(x, y, z) = x^2 y + x y z$$

$$= (\theta x)^2 (\theta y) + \theta x \theta y \theta z$$

$$= \theta^3 x^2 y + \theta^3 x y z$$

$$= \theta^3 [x^2 y + x y z]$$

Homogenous if degree - 3

▣ Properties of homogenous functions:-

if $f(x_i)$ is homogenous of degree K in x_i 's

then, $\frac{\partial f}{\partial x_i}$ will be held ~~K~~ $(K-1)$ in x_i 's

x_i 's

??

e.g., CRTS technology $q = \sqrt{KL}$

$$\frac{\partial q}{\partial L} = \frac{1}{2} \cdot \sqrt{\frac{K}{L}}$$

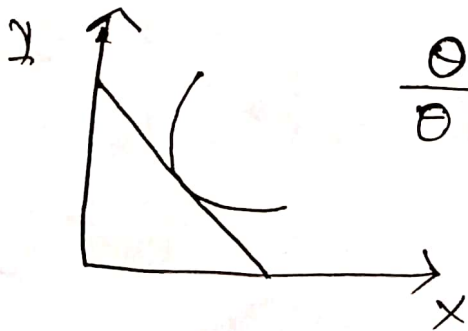
$$\Rightarrow \frac{\partial q}{\partial K} = \frac{1}{2} \sqrt{\frac{L}{K}}$$

h.d = homogenous degree

if $f(x)$ is h.d. K and $g(x)$ is h.d. m then

$f(x) \cdot g(x)$ is h.d. $(K+m)$

$f(x)/g(x)$ is h.d. $(K-m)$



$$\frac{\partial I}{\partial P_y} = -\frac{I}{P_y}$$

□ Eulers law:-

Let $f(x)$ be h.d $\propto M$ i.e x

Then $\sum_{i=1}^n \frac{\partial f}{\partial x_i}, x_i = x f(x)$

go back to $U = xy$

$$\frac{\partial U}{\partial x} = y ; x \frac{\partial U}{\partial x} = xy$$

$$\frac{\partial U}{\partial y} = x ; y \frac{\partial U}{\partial y} = xy$$

$$x \frac{\partial U}{\partial x} + y \frac{\partial U}{\partial y} = 2xy = 2U(x, y)$$

(homogenous h.d - 2)

$$\hat{U}(x, y) = f(U(x, y)) \text{ where } f' > 0 \text{ and}$$

$$U(x, y) \text{ h.d, } I$$



$$MRS = \frac{\partial U / \partial x}{\partial U / \partial y}$$

$$= \frac{\partial f / \partial y}{\partial f / \partial x} = \frac{\partial y / \partial x}{\partial y / \partial x}$$

$$= \frac{\partial y / \partial x}{\partial y} = MRS = f\left(\frac{y}{x}\right)$$

but MRS is h.d.o (i.e. a function) if $\frac{y}{x}$

$$f\left(\frac{y}{x}\right) = \frac{p_x}{p_y} \cdot \frac{y}{x} = f^{-1}\left(\frac{p_x}{p_y}\right)$$

$y = \text{constant} \cdot x$

Primal and Dual are related -
अपना एक indirect utility - reverse

1 October - 2024

L-05

Primal Vs Dual

Primal problem in consumer theory -

$$\text{Max } U = U(x, y)$$

$$\text{s.t } P_x x + P_y y = I$$

From this problem, we got Marshallian demands,

Simple Case!

$$U = xy$$

Primal Problem

$$\text{Max}_{x, y, \lambda} \mathcal{L} = xy + \lambda [I - P_x x - P_y y]$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= y - \lambda P_x = 0 \\ \frac{\partial \mathcal{L}}{\partial y} &= x - \lambda P_y = 0 \end{aligned} \quad \left\{ \begin{aligned} \frac{y}{x} &= \frac{P_x}{P_y} \Rightarrow y = \frac{P_x x}{P_y} \\ y &= \frac{P_x}{P_y} \cdot \frac{I}{2P_x} \\ y^* &= \frac{I}{2P_y} \end{aligned} \right.$$

$$\frac{\partial \mathcal{L}}{\partial x} = I - P_x x - P_y y = 0$$

$$\Rightarrow I = P_x x + P_y y$$

$$\Rightarrow I = P_x x + P_y \cdot \frac{P_x x}{P_y}$$

$$\Rightarrow I = 2P_x x \Rightarrow x^* = \frac{I}{2P_x}$$

Let's plug the demands back in to Lagrange:

$$h^*(x^*, y^*, I, P_x, P_y) = x^* y^* + \lambda (I - P_x x^* - P_y y^*)$$

$$= \frac{I}{2P_x} \cdot \frac{I}{2P_y} + \lambda \left(I - P_x \cdot \frac{I}{2P_x} - P_y \cdot \frac{I}{2P_y} \right)$$

$$\therefore \mathcal{L}^*(x^*, y^*, I, P_x, P_y) = \frac{I^2}{4P_x P_y} = U(x(P, I), y(P, I))$$

$$U(x^*(P, I), y^*(P, I)) \equiv V(P, I) \equiv \text{Indirect utility function}$$

Indirect Utility: for a given set of prices and income gives the highest attainable utility

$P =$
small
 P

Our example: $v(p, I) = \frac{I^2}{4p_x p_y}$

Can we get demands from v ? Envelope theorem:

$$v = L^*(x^*(p, I), y^*(p, I), p, I) = U(x^*(p, I), y^*(p, I)) + \lambda (I - p_x x^* - p_y y^*)$$

$$\frac{\partial v}{\partial p_x} = \frac{\partial L^*}{\partial p_x} = \frac{\partial L^*}{\partial x} \cdot \frac{\partial x}{\partial p_x} + \frac{\partial L^*}{\partial y} \cdot \frac{\partial y}{\partial p_x} - \lambda x^*$$

$$\frac{\partial v}{\partial p_x} = -\lambda x^*(p, I)$$

$$\frac{\partial v}{\partial I} = \frac{\partial L^*}{\partial I} = \frac{\partial L^*}{\partial x} \cdot \frac{\partial x}{\partial I} + \frac{\partial L^*}{\partial y} \cdot \frac{\partial y}{\partial I} + \lambda$$

$$\frac{\partial v}{\partial I} = \lambda$$

$$\frac{\lambda_x}{\lambda} = - \frac{\partial v / \partial p_x}{\partial v / \partial I}$$

$$\Rightarrow \lambda_x = - \frac{\partial v / \partial p_x}{\partial v / \partial I} \quad \text{Roy's identity}$$

Our example: $v = \frac{I^2}{4P_X P_Y} \Rightarrow -\frac{\partial v / \partial P_X}{\partial v / \partial I} = -4 \frac{I^2}{4P_X^2 P_Y}$

$$\frac{\partial v}{\partial I} = \frac{2 \cdot I}{4P_X P_Y}$$

$$= \frac{I}{2P_X}$$

Primal problem:-

$$\text{Max } U(X, Y)$$

$$\text{s.t. } P_X X + P_Y Y = I$$

$$v(P, I) = U(X^*(P, I), Y^*(P, I))$$

Roy's identity
 $X^* = -\frac{\partial v / \partial P_X}{\partial v / \partial I}, Y^* = -\frac{\partial v / \partial P_Y}{\partial v / \partial I}$

Marshallian demand fⁿ
 $X^*(P, I)$
 $Y^*(P, I)$

Primal Problem जहाँ Roy's identity का प्रयोग
 Ans. - ~~is~~

Dual Problem :-

$$\text{Min } P_x X + P_y Y$$

$$\text{s.t. } U = U(X, Y) = \bar{U}$$

$$\text{Min}_{X, Y, \lambda} \mathcal{L} = P_x X + P_y Y + \lambda [\bar{U} - U(X, Y)]$$

$$\left. \begin{aligned} \frac{\partial \mathcal{L}}{\partial X} &= P_x - \lambda \frac{\partial U}{\partial X} = 0 \\ \frac{\partial \mathcal{L}}{\partial Y} &= P_y - \lambda \frac{\partial U}{\partial Y} = 0 \end{aligned} \right\} \frac{P_x/P_y}{\partial U/\partial X / \partial U/\partial Y} = \frac{\partial U/\partial X}{\partial U/\partial Y} = \text{MRS}_{XY}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \bar{U} - U(X, Y) = 0$$

$$\bar{U} = U(X, Y) = \text{constant utility}$$

$$U = XY$$

$$\frac{Y}{X} = \frac{P_x}{P_y} \quad \text{Tangency}$$

$$Y = \frac{P_x}{P_y} X$$

$$\bar{U} = X \cdot X \cdot \frac{P_x}{P_y} = X^2 \frac{P_x}{P_y}$$

$$X^2 = \bar{U} \cdot \frac{P_y}{P_x}$$

$$X = \left(\bar{U} \cdot \frac{P_y}{P_x} \right)^{1/2}$$

$= h_x(p, \bar{U}) =$ Hicksian demand
compensated demand

$$h_x(p, \bar{U}) = \left(\frac{\bar{U} P_y}{P_x} \right)^{1/2}$$

$$\frac{P_x}{P_y} = \frac{y}{x} \Rightarrow y = \frac{P_x x}{P_y} = \frac{P_x}{P_y} \left(\frac{\bar{U} P_y}{P_x} \right)^{1/2}$$

$$h_y(p, \bar{U}) = \left(\frac{\bar{U} P_x}{P_y} \right)^{1/2}$$

$$\min_{x, y, \lambda} \mathcal{L} = P_x x + P_y y + \lambda [\bar{U} - u(x, y)]$$

Optimal value function:

$$\mathcal{L}^* = P_x h_x(p, \bar{U}) + P_y h_y(p, \bar{U}) + \lambda [\bar{U} - u(h_x(p, \bar{U}), h_y(p, \bar{U}))]$$

$$\mathcal{L}^* = P_x h_x(p, \bar{U}) + P_y h_y(p, \bar{U}) = e(p, \bar{U})$$

Expenditure function:

Tells the minimum cost of a bundle you like as much as of given bundle for a set of prices.

$$e(p, \bar{u}) = P_x \left(\frac{\bar{u} P_x}{P_x} \right)^{1/2} + P_y \left(\frac{\bar{u} P_y}{P_y} \right)^{1/2}$$

$$e(p, \bar{u}) = 2 \sqrt{\bar{u} P_x P_y}$$

$$e(p, \bar{u}) = \mathcal{L}^*(n_x, n_y, p, \bar{u})$$

$$* e(p, \bar{u}) \mathcal{L}^* = P_x h_x(p, \bar{u}) + P_y h_y(p, \bar{u}) + \lambda (\bar{u} - u(n_x, n_y))$$

$$\frac{\delta \mathcal{L}}{\delta P_x} = \underbrace{\left(\frac{\delta \mathcal{L}}{\delta h_x} \right)}_{=0} \frac{\delta h_x}{\delta P_x} + \underbrace{\left(\frac{\delta \mathcal{L}}{\delta h_y} \right)}_{=0} \frac{\delta h_y}{\delta P_x} + \underbrace{\frac{\delta \mathcal{L}}{\delta P_x}}_{=h_x}$$

$$h_x(p, \bar{u}) = \frac{\delta e}{\delta P_x} \quad h_y = \frac{\delta e}{\delta P_y}$$

$$\boxed{\text{Min } P_x X + P_y Y}$$

$$\text{s.t. } U \geq U(x, y) = \bar{U}$$

$$e(p, \bar{U}) = \mathcal{L}^* (h_x, h_y, p, \bar{U})$$

Shephard's Lemma

$$\frac{\partial e(p, \bar{U})}{\partial P_x} = h_x$$

$$\frac{\partial e(p, \bar{U})}{\partial P_y} = h_y$$

Substitution

$$h_x(p, \bar{U})$$

$$h_y(p, \bar{U})$$

Lemma states that if IC's of expenditure or cost fn are convex, then the cost minimizing point of a given good (i) ~~with~~ with price P_i is unique.

Ex $\Rightarrow$ min $\Rightarrow$ utility maximize.